

signals

Unix-like *operating system feature*

like exceptions, but for processes:

- can be triggered by an external process
 - kill command/system call

- can be triggered by special events
 - pressing control-C
 - other events that would normally terminate program
 - 'segmentation fault'
 - illegal instruction
 - divide by zero

- can invoke *signal handler* (like exception handler)

exceptions v signals

(hardware) exceptions	signals
handler runs in kernel mode	handler runs in user mode
hardware decides when	OS decides when
hardware needs to save PC	OS needs to save PC + registers
processor program counter changes	thread program counter changes
program counter = instruction to run next	

exceptions v signals

(hardware) exceptions	signals
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hardware decides when	<i>OS decides when</i>
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...but OS needs to run to trigger handler
most likely “forwarding” hardware exception

exceptions v signals

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hardware decides when	OS decides when
hardware needs to save PC	OS needs to save PC + <i>registers</i>
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program counter = instruction to run next	

signal handler follows normal calling convention
not special assembly like typical exception handler

exceptions v signals

(hardware) exceptions	signals
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hardware decides when	OS decides when
hardware needs to save PC	OS needs to save PC + registers
<i>processor</i> program counter changes	<i>thread</i> program counter changes
program counter = instruction to run next	

signal handler runs in same thread ('virtual processor')
as process was using before

not running at 'same time' as the code it interrupts

base program

```
int main() {  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read_%s", buf);  
    }  
}
```

base program

```
int main() {  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read %s", buf);  
    }  
}
```

some input

read some input

more input

read more input

(control-C pressed)

(program terminates immediately)

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new program

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int main() {  
    ... // added stuff shown later  
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    ... // added stuff shown later  
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    while (fgets(buf, sizeof buf, stdin)) {  
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    }  
}
```

some input

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read more input

(control-C pressed)

Control-C pressed?!

another input **read another input**

example signal program

```
void handle_sigint(int signum) {  
    /* signum == SIGINT */  
    write(1, "Control-C\ncpress?!\n", sizeof("Control-C\ncpress?!\n"));  
}  
  
int main(void) {  
    struct sigaction act;  
    act.sa_handler = &handle_sigint;  
    sigemptyset(&act.sa_mask);  
    // SA_RESTART = if syscall interrupted,  
    // complete it when handler returns  
    act.sa_flags = SA_RESTART;  
    sigaction(SIGINT, &act, NULL);  
  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read\nc%s", buf);  
    }  
}
```

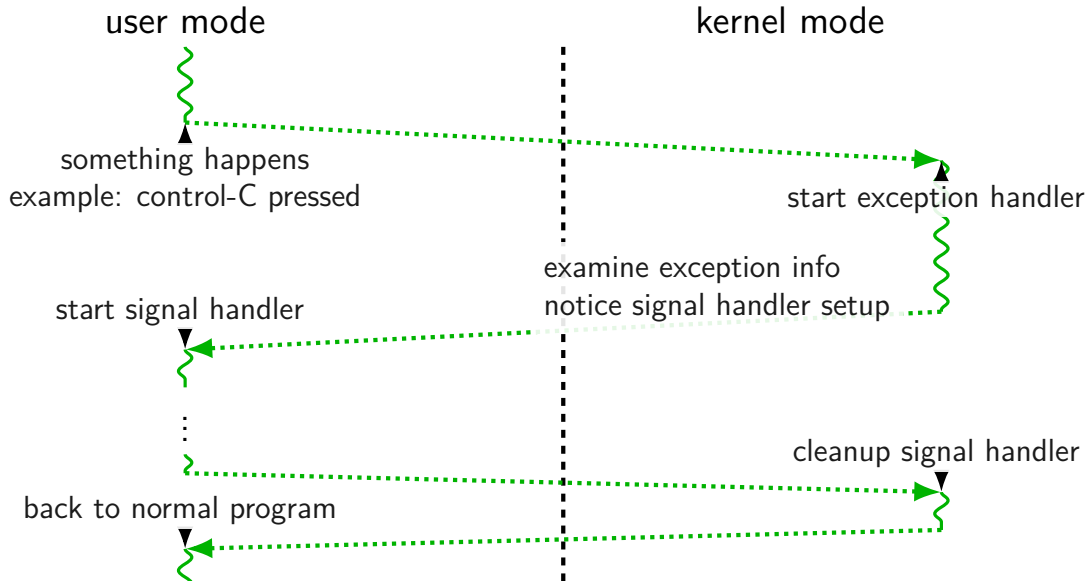
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    // SA_RESTART = if syscall interrupted,  
    // complete it when handler returns  
    act.sa_flags = SA_RESTART;  
    sigaction(SIGINT, &act, NULL);  
  
    char buf[1024];  
    while (fgets(buf, sizeof buf, stdin)) {  
        printf("read\nc%s", buf);  
    }  
}
```

'forwarding' exception as signal



SIGxxxx

signals types identified by number...

constants declared in `<signal.h>`

constant	likely use
SIGBUS	"bus error"; certain types of invalid memory accesses
SIGSEGV	"segmentation fault"; other types of invalid memory accesses
SIGINT	what control-C usually does
SIGFPE	"floating point exception"; includes integer divide-by-zero
SIGHUP, SIGPIPE	reading from/writing to disconnected terminal/socket
SIGUSR1, SIGUSR2	use for whatever you (app developer) wants
SIGKILL	terminates process (cannot be handled by process!)
SIGSTOP	suspends process (cannot be handled by process!)
...	...

SIGxxxx

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SIGKILL	terminates process (<i>cannot be handled by process!</i>)
SIGSTOP	suspends process (<i>cannot be handled by process!</i>)
...	...

handling Segmentation Fault

```
...  
void handle_sigsegv(int num) {  
    puts("got SIGSEGV");  
}  
  
int main(void) {  
    struct sigaction act;  
    act.sa_handler = handle_sigsegv;  
    sigemptyset(&act.sa_mask);  
    act.sa_flags = SA_RESTART;  
    sigaction(SIGSEGV, &act, NULL);  
  
    asm("movq %rax, 0x12345678");  
}
```

handling Segmentation Fault

```
...  
void handle_sigsegv(int num) {  
    puts("got SIGSEGV");  
}  
  
int main(void) {  
    struct sigaction act;  
    act.sa_handler = handle_sigsegv;  
    sigemptyset(&act.sa_mask);  
    act.sa_flags = SA_RESTART;  
    sigaction(SIGSEGV, &act, NULL);  
  
    asm("movq %rax, 0x12345678");  
}
```

got SIGSEGV

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signal API

`sigaction` — register handler for signal

`kill` — send signal to process

uses *process ID* (integer, retrieve from `getpid()`)

`pause` — put process to sleep until signal received

`sigprocmask` — temporarily block/unblock some signals from being received

signal will still be *pending*, received if unblocked

... and much more

kill command

kill command-line command : calls the kill() function

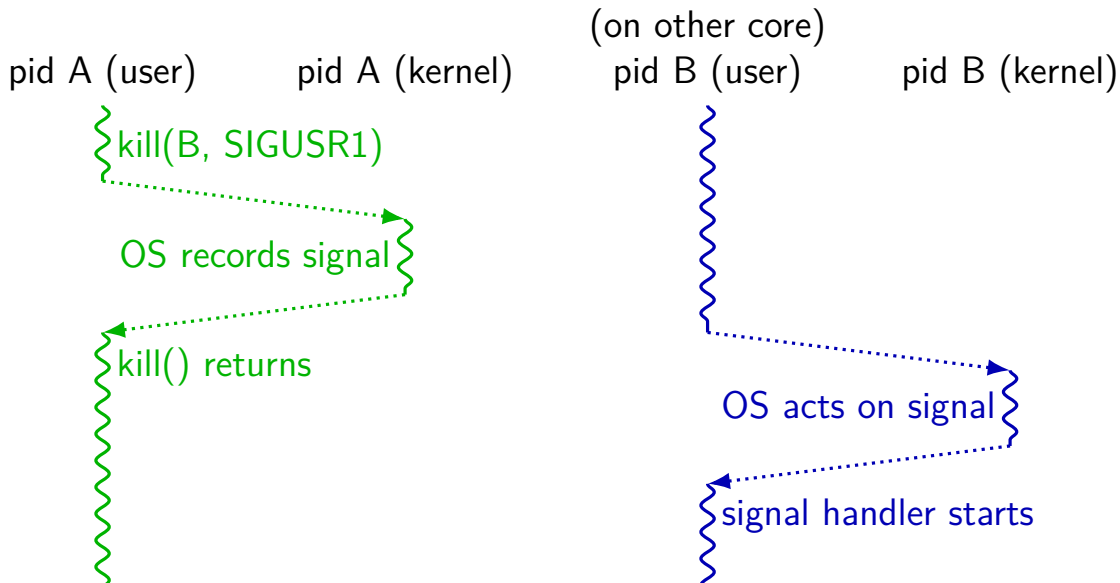
`kill 1234` — sends SIGTERM to pid 1234

in C: `kill(1234, SIGTERM)`

`kill -USR1 1234` — sends SIGUSR1 to pid 1234

in C: `kill(1234, SIGUSR1)`

kill() not always immediate



output of this?

pid 1000

pid 2000



```
void handle_usr1(int num) {  
    write(1, "Y", 1);  
    _exit(0);  
}  
  
int main() {  
    struct sigaction act;  
    ... // initialize rest of "act"  
    act.sa_handler = &handle_usr1;  
    sigaction(SIGUSR1, &act, NULL);  
}
```

If these run at same time, expected output?

- A. XY
- B. X
- C. Y
- D. YX
- E. X or XY, depending on timing
- F. crash
- G. (nothing)
- H. something else

output of this? (v2)

pid 1000

pid 2000



```
void handle_usr1(int num) {
    write(1, "Y", 1);
    _exit(0);
}
int main() {
    struct sigaction act;
    ... // initialize rest of "act"
    act.sa_handler = &handle_usr1;
    sigaction(SIGUSR1, &act);
    while (1) pause(); // ADDED
}
```

If these run at same time, expected output?

- A. XY
- B. X
- C. Y
- D. YX
- E. X or XY, depending on timing
- F. crash
- G. (nothing)
- H. something else

signal handler unsafety (0)

```
void foo() {  
    /* SIGINT might happen while foo() is running */  
    char *p = malloc(1024);  
    ...  
}  
  
/* signal handler for SIGINT  
(registered elsewhere with sigaction()) */  
void handle_sigint() {  
    printf("You pressed control-C.\n");  
}
```

signal handler unsafety (1)

```
void *malloc(size_t size) {  
    ...  
    to_return = next_to_return;  
    /* SIGNAL HAPPENS HERE */  
    next_to_return += size;  
    return to_return;  
}  
  
void foo() {  
    /* This malloc() call interrupted */  
    char *p = malloc(1024);  
    p[0] = 'x';  
}  
  
void handle_sigint() {  
    // printf might use malloc()  
    printf("You pressed control-C.\n");  
}
```

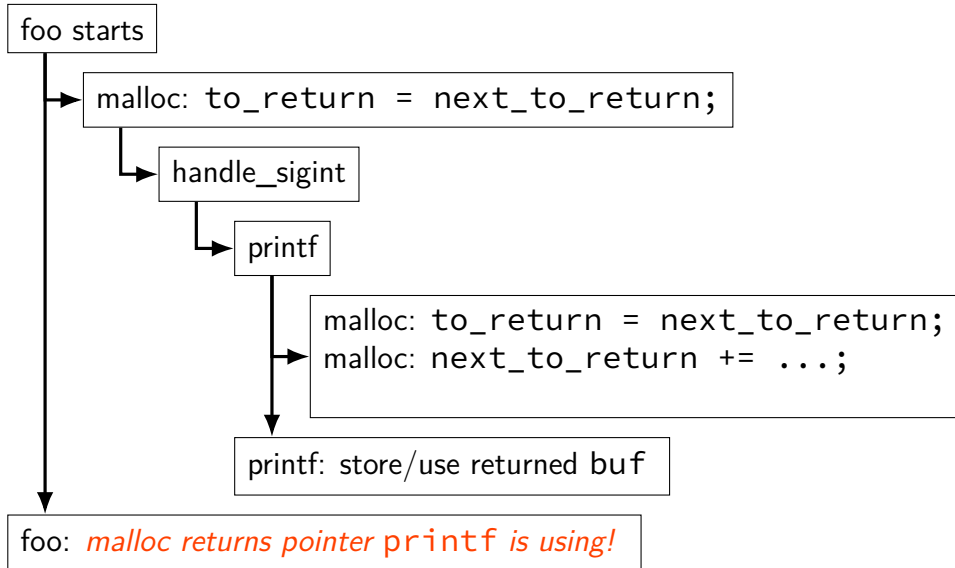
signal handler unsafety (1)

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void *malloc(size_t size) {  
    ...  
    to_return = next_to_return;  
    /* SIGNAL HAPPENS HERE */  
    next_to_return += size;  
    return to_return;  
}  
  
void foo() {  
    /* This malloc() call interrupted */  
    char *p = malloc(1024);  
    p[0] = 'x';  
}  
  
void handle_sigint() {  
    // printf might use malloc()  
    printf("You pressed control-C.\n");  
}
```

signal handler unsafety (2)

```
void handle_sigint() {  
    printf("You pressed control-C.\n");  
}  
  
int printf(...) {  
    static char *buf;  
    ...  
    buf = malloc()  
    ...  
}
```

signal handler unsafety: timeline



signal handler unsafety (3)

```
foo() {  
    char *p = malloc(1024)... {  
        to_return = next_to_return;  
        handle_sigint() { /* signal delivered here */  
            printf("You pressed control-C.\n") {  
                buf = malloc(...) {  
                    to_return = next_to_return;  
                    next_to_return += size;  
                    return to_return;  
                }  
                ...  
            }  
        }  
        next_to_return += size;  
        return to_return;  
    }  
    /* now p points to buf used by printf! */  
}
```

signal handler unsafety (3)

```
foo() {  
    char *p = malloc(1024)... {  
        to_return = next_to_return;  
        handle_sigint() { /* signal delivered here */  
            printf("You pressed control-C.\n") {  
                buf = malloc(...) {  
                    to_return = next_to_return;  
                    next_to_return += size;  
                    return to_return;  
                }  
                ...  
            }  
        }  
        next_to_return += size;  
        return to_return;  
    }  
    /* now p points to buf used by printf! */  
}
```

signal handler safety

POSIX (standard that Linux follows) defines “async-signal-safe” functions

`man signal-safety` can get you list on portal

these must work correctly no matter what they interrupt

...and no matter how they are interrupted

includes: `write`, `_exit`

does not include: `printf`, `malloc`, `exit`

blocking signals

avoid having signal handlers anywhere:

can instead *block signals*

`sigprocmask()`, `pthread_sigmask()`

blocked = signal handler doesn't run

also called “masking” a signal

blocked signals are not *delivered* (acted on)

instead, signal becomes *pending*

delivered if unblocked

blocking signals

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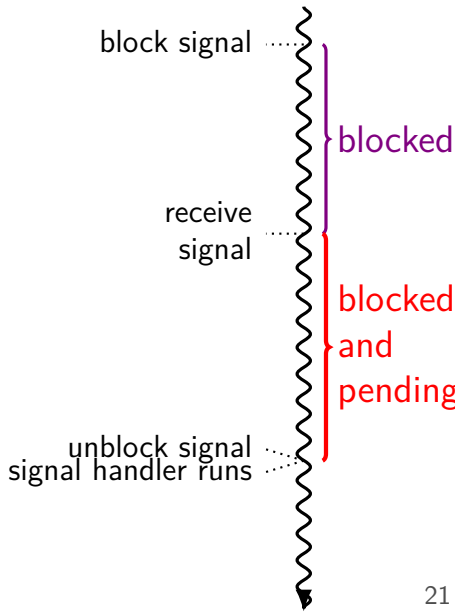
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controlling when signals are handled

first, block a signal

then either unblock signals only at certain times

some special functions to help:

`sigsuspend` (unblock and wait until handler runs),

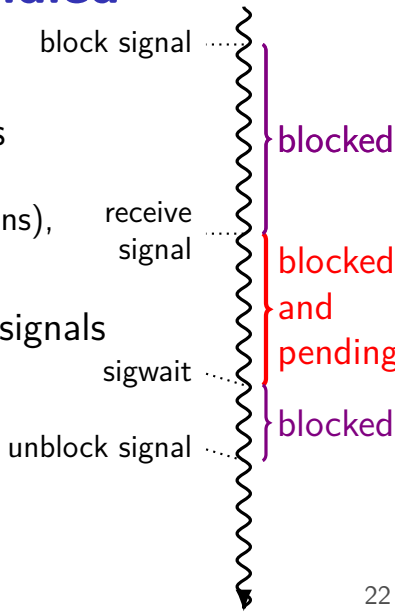
`pselect` (unblock while checking for I/O), ...

and/or use API for checking/changing pending signals

example: `sigwait`

(wait for blocked signal to become pending;
then mark not pending)

typically *instead of having signal handler*



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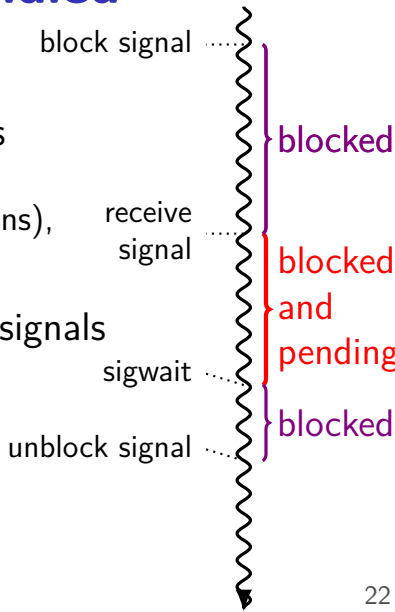
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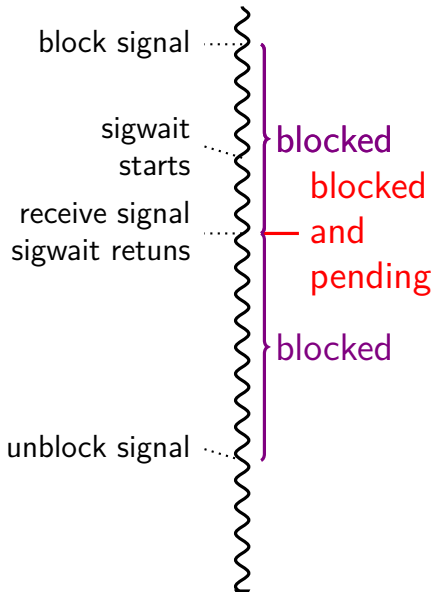
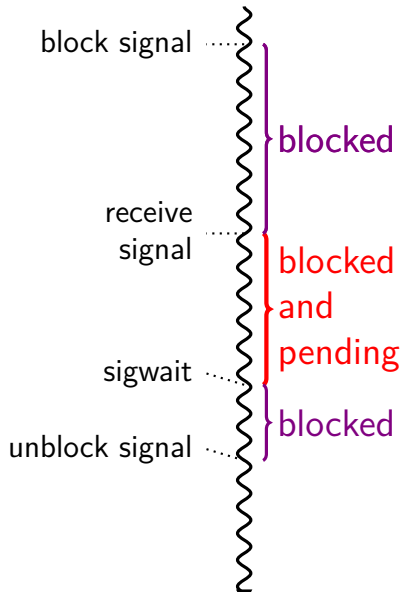
example: `sigwait`

(wait for blocked signal to become pending;
then mark not pending)

typically *instead of having signal handler*



sigwait timelines



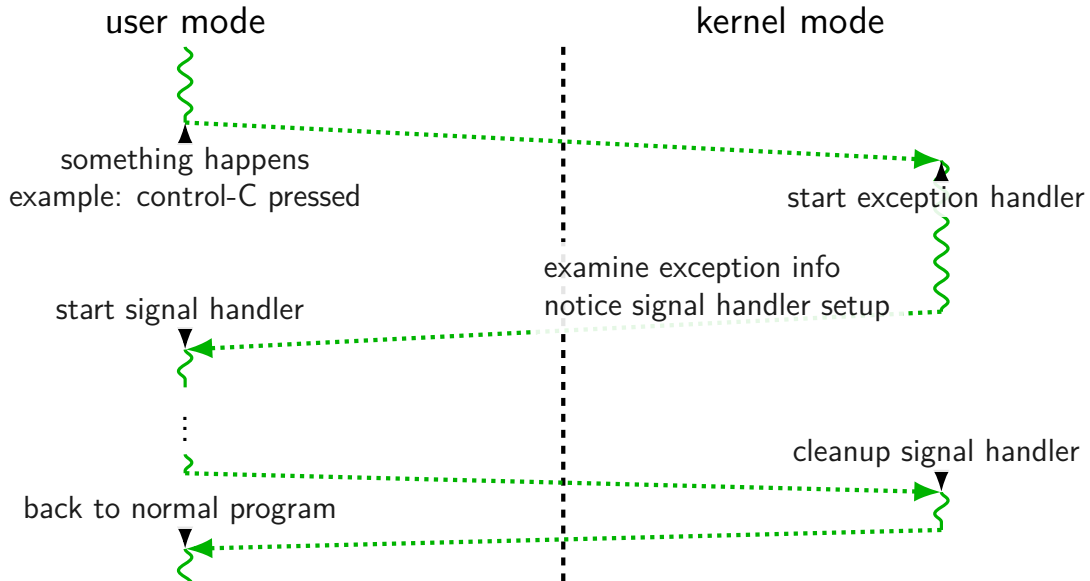
synchronous signal handling

```
int main(void) {
    sigset_t set;
    sigemptyset(&set);
    sigaddset(&set, SIGINT);
    sigprocmask(SIG_BLOCK, &set, NULL);

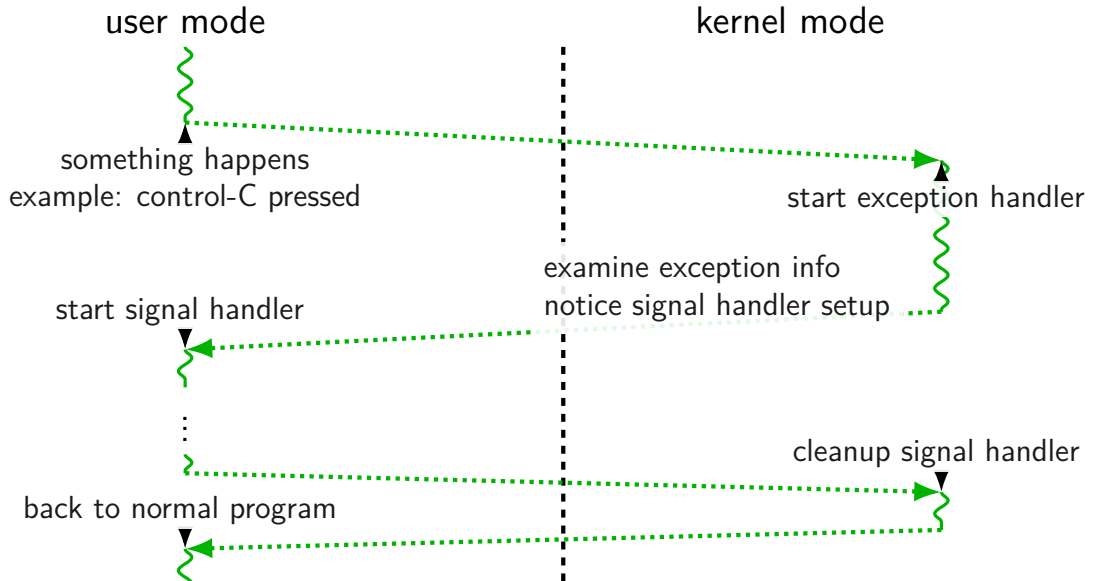
    printf("Waiting for SIGINT (control-C)\n");
    int num;
    if (sigwait(&set, &num) != 0) {
        printf("sigwait failed!\n");
    }
    if (num == SIGINT);
        printf("Got SIGINT\n");
    }
}
```

backup slides

'forwarding' exception as signal



'forwarding' exception as signal



x86-64 Linux signal delivery (1)

suppose: signal (with handler) happens while `foo()` is running

should stop in the middle of `foo()`

do signal handler

go back to `foo()` without...

changing local variables (possibly in registers)

(and `foo()` doesn't have code to do that)

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x86-64 Linux signal delivery (2)

suppose: signal (with handler) happens while `foo()` is running

OS saves registers *to user stack*

OS modifies user registers, PC to call signal handler

the stack

address of <code>__restore_rt</code>
saved registers
PC when signal happened
local variables for <code>foo</code>
...

→ stack pointer
when signal handler started

→ stack pointer
before signal delivered

x86-64 Linux signal delivery (3)

```
handle_sigint:
```

```
...
```

```
ret
```

```
...
```

```
__restore_rt:
```

```
    // 15 = "sigreturn" system call
```

```
    movq $15, %rax
```

```
    syscall
```

__restore_rt is *return address* for signal handler

sigreturn syscall restores pre-signal state

- if SA_RESTART set, restarts interrupted operation

- also handles caller-saved registers

- also might change which signals blocked (depending how sigaction was called)

SA_RESTART

```
struct sigaction sa; ...  
sa.sa_flags = SA_RESTART;
```

general version:

```
sa.sa_flags = SA_NAME | SA_NAME | SA_NAME; (or 0)
```

if SA_RESTART included:

after signal handler runs, attempt to restart interrupted operations (e.g. reading from keyboard)

if SA_RESTART not included:

after signal handler runs, interrupted operations return typically an error (detect by checking `errno == EINTR`)

sending signals (1)

PID 1000:

```
void handle_usr1(int num) {
    write(1, "Y", 1);
    kill(2000, SIGUSR2);
}

int main() {
    struct sigaction act;
    ... // initialize act
    act.sa_handler = &handle_usr1;
    sigaction(SIGUSR1, &act, NULL);
    // wait for pid 2000 to start
    sleep(60);
    kill(2000, SIGUSR1);
    while (1) pause();
}
```

sending signals (2)

PID 1000:

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void handle_usr1(int num) {
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    sigaction(SIGUSR1, &act, NULL);
    sleep(60); // wait for pid 2000
    kill(2000, SIGUSR1);
    while (1) pause();
}
```

PID 2000:

```
void handle_usr1(int num) {
    write(1, "X", 1);
    kill(1000, SIGUSR1);
}
```

```
void handle_usr2(int num) {
    write(1, "Z", 1);
    kill(1000, SIGTERM);
    _exit(0);
}
```

```
int main() {
    struct sigaction act;
    ... // initialize act
    act.sa_handler = &handle_usr1;
    sigaction(SIGUSR1, &act, NULL);
    act.sa_handler = &handle_usr2;
    sigaction(SIGUSR2, &act, NULL);
    while (1) pause();
}
```

sending signals (2)

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    write(1, "Y", 1);  
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}
```

```
int main() {  
    struct sigaction act;  
    act.sa_handler = &handle_usr1;  
    sigaction(SIGUSR1, &act, NULL);  
    act.sa_handler = &handle_usr2;  
    sigaction(SIGUSR2, &act, NULL);  
    while (1) pause();  
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sending signals (2)

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sending signals (2)

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    kill(2000, SIGUSR1);  
    while (1) pause();  
}
```

PID 2000:

```
void handle_usr1(int num) {  
    write(1, "X", 1);  
    kill(1000, SIGUSR1);  
}
```

```
void handle_usr2(int num) {  
    write(1, "Z", 1);  
    kill(1000, SIGTERM);  
    _exit(0);  
}
```

```
int main() {  
    struct sigaction act;  
    act.sa_handler = &handle_usr1;  
    sigaction(SIGUSR1, &act, NULL);  
    act.sa_handler = &handle_usr2;  
    sigaction(SIGUSR2, &act, NULL);  
    while (1) pause();  
}
```